TO SAVE THE BEST MODEL

⇓

joblib.dump(best_pipeline, "mental_health_pipeline.pkl")

BUILDING FAST-API PYDANTIC BACKEND

1. import joblib

2. LOAD THE MODEL

↳ model = joblib.load("Mental_health_model.pkl")

③ TO RUN A FAST-API SERVER

```
python -m unicorn main:app --reload.
```

④ CREATE A POST ROUTE

```
@app.post('predict)
def predict( ):
```

⑤ CREATE A PYDANTIC MODEL

```
class student data(BaseModel):
    age: int
    gender: str
    :
    :
    stress_level: str
```

⑥ USE THE ABOVE MODEL AS THE REQ BODY OF /predict route.

⑦ `prediction = model.predict(input_row)[0]`

DEEP LEARNING

When Traditional learning ends, Deep Learning begins turning raw data into intelligence, and intelligence into discoveries.

What is Deep Learning?

Imagine your brain has lots of tiny switches (neurons)
When you see a cat, your brain looks at:

- the shape of the ears.
- The whiskers.
- The fur
- The sound "meow".

Deep Learning works the same way. It builds layers of tiny switches (artificial neurons) inside a computer. Each layer learns something:

- First layer: "This looks like an edge".
- Next layer: "This looks like an ear".
- Final layer: "Oh! It's a cat."

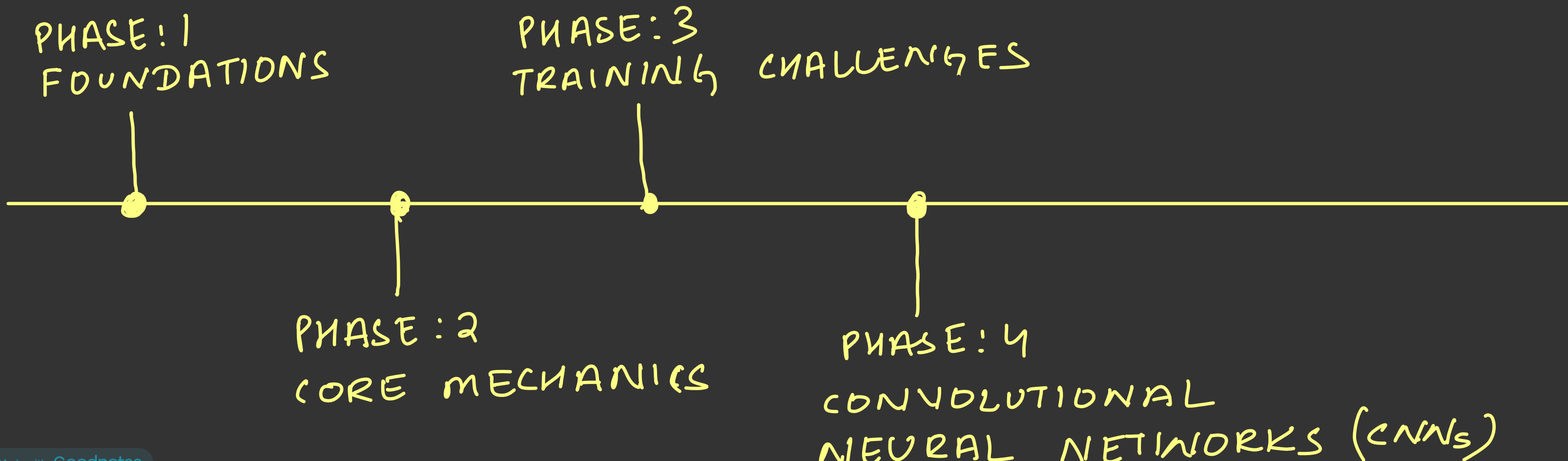
Learning from mistakes until it gets really good.

WHY IS DL POPULAR NOW?

- Data explosion
- Hardware power (GPUs & TPUs)
  - ↳ Training deep networks needs a lot of calculations.

↳ Earlier computers were slow, now we have GPUs & TPUs.

- Automatic feature learning.
- Better performance.
- Versatility Across Domains.
- Research Breakthroughs.



PERCEPTRON

The perceptron is the simplest type of artificial neuron. It is the foundation of neural network and deep learning.

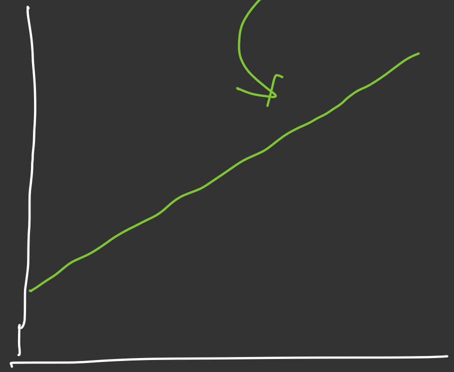
It mimics, how a biological neuron works taking inputs, processing them and giving output.

Table $\rightarrow$

Age	Cholestrol	BP	heart_disease
28	150	110	1
36	120	90	0
42	~~120~~ 180	160	1

step-1 $\Rightarrow$ $y = mx + b \rightarrow h_\theta(x) = \theta_0 + \theta_1 x_1$

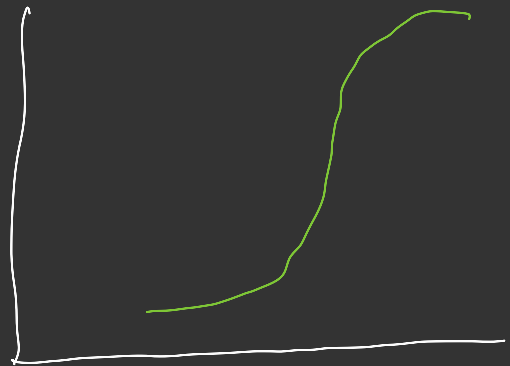
$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$



STEP-2

SIGMOID ACTIVATION FUNCTION

$$= \frac{1}{1+e^{-z}} \Rightarrow z = (\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$$



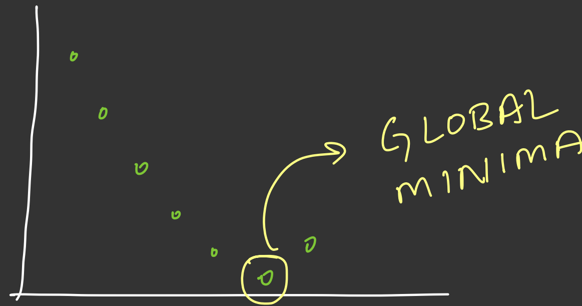
STEP-3

Log loss function

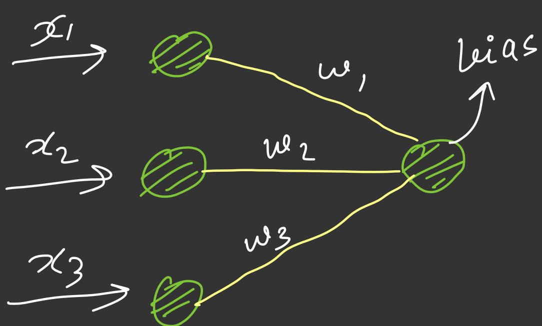
$\Rightarrow$ loss find.

$\boxed{\text{STEP-4}}$ - OPTIMIZER

$$\theta_0 + \theta_1 x_1$$



we did all of these using PERCEPTRONS.



$$x_1 \times w_1 + x_2 \times w_2 + x_3 w_3 + \text{bias}$$

$$(\theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_0)$$

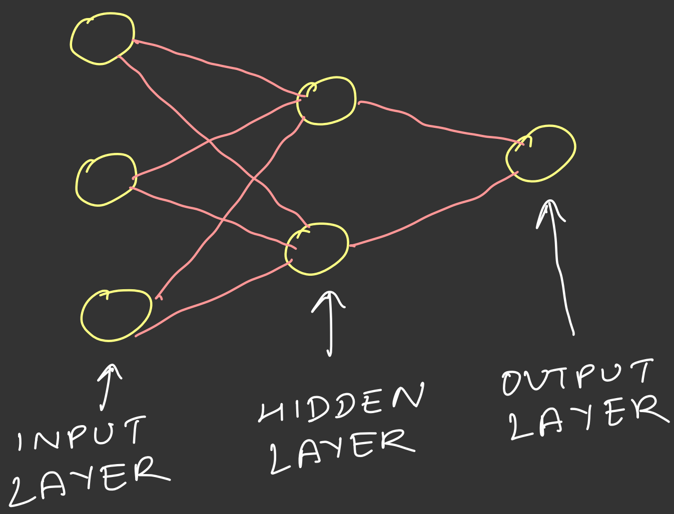
same sigmoid activation function.

$$= \frac{1}{1+e^{(-2)}}$$

$\boxed{\text{LIMITATIONS OF PERCEPTRONS}}$

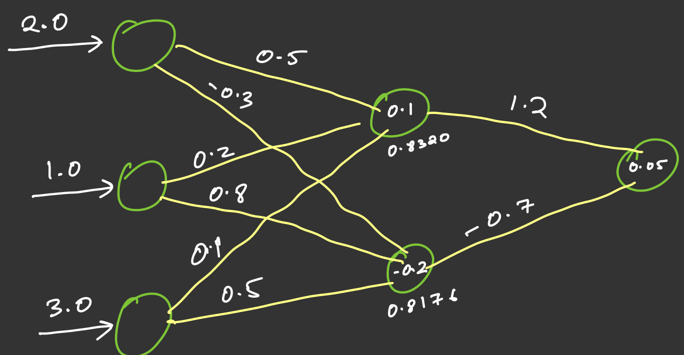
↳ step activation function → 0/1

ANN (Artificial Neural Network)



FORWARD PROPAGATION

x_1	x_2	x_3	y
2.0	1.0	3.0	1



$$hn_1 = (x_1 w_1) + (x_2 w_2) + (x_3 w_3) + bias$$

$$= (2.0)(0.5) + (1.0)(0.2) + (3.0)(0.1)$$

$$\approx 1.6$$

Made with Goodnotes

$$\sigma = \frac{1}{1+e^{-z}} \qquad \sigma(1.6) = 0.8320$$

output neuron $= [0.8320\ (1.2)] + [0.8176\ (0.7)]$
$+ 0.05$

$= 0.4761$

$\sigma(0.4761) = 0.6168 \quad > 0.5$

≈ 1

$$\boxed{y\text{-}pred = 1}$$

★ 3 IMPORTANT THINGS

ACTIVATION FUNCTION — LOSS FUNCTION — OPTIMIZERS

forward propagation

Backward propagation

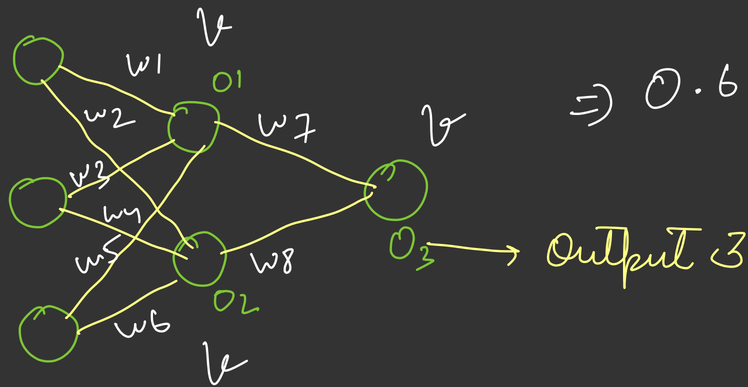
BACKWARD PROPAGATION

$\hat{y} = 0.6168$

$y = 1$

$$\text{Log loss} = [y \times \log(\hat{y}) + (1-y) \times \log(1-\hat{y})$$

$$= 0.494$$



$\Rightarrow 0.6168$

LOSS = 0.494 ↓

① UPDATE THE WEIGHTS

$$w_{new} = w_{old} - \eta \frac{dL}{dw_{old}}$$

learning rate

L = Loss

$$w_{7\,new} = w_{7\,old} - \eta \left[\frac{dL}{dw_{7\,old}} \right] \Rightarrow \frac{dL}{dO_3} \times \frac{dO_3}{dw_{7\,old}}$$

$$w_{1}\,\text{new} = w_{1}\,\text{old} - \eta\left[\frac{dL}{dw_{1}\,\text{old}}\right] \Rightarrow \frac{dL}{dO_3} \times \frac{dO_3}{dO_1} \times \frac{dO_1}{dw_{1}\,\text{old}}$$

$$w_{6}\,\text{new} = w_{6}\,\text{old} - \eta\left(\quad\right) \qquad \frac{dL}{dw_{6}\,\text{old}} = \frac{dL}{dO_3}$$

Made with Goodnotes